

WASP-43b

R-1655

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220127 · created 2026-08-02T08:35:37+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459606.8153 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.015
Transit depth (Rp/Rs)^2	3.53 +/- 0.55 [%]
Orbital Inclination (inc)	81.08 +/- 1.22
Airmass coefficient 1 (a1)	0.9853 +/- 0.0094
Airmass coefficient 2 (a2)	0.007 +/- 0.016
Scatter in the residuals of the lightcurve fit is	0.96 %
Best Comparison Star	#2 - [203, 110]
Optimal Method	PSF photometry
Transit Duration (day)	0.0477 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.188 ± 0.015 vs 0.1594 (deviation 1.91σ)
Duration fit vs published	0.0477 d vs 0.0512 d (deviation 0.93σ)
Mid-time O–C	1.6 min
Depth SNR	12.5
Residual scatter	0.96 %

Light curve

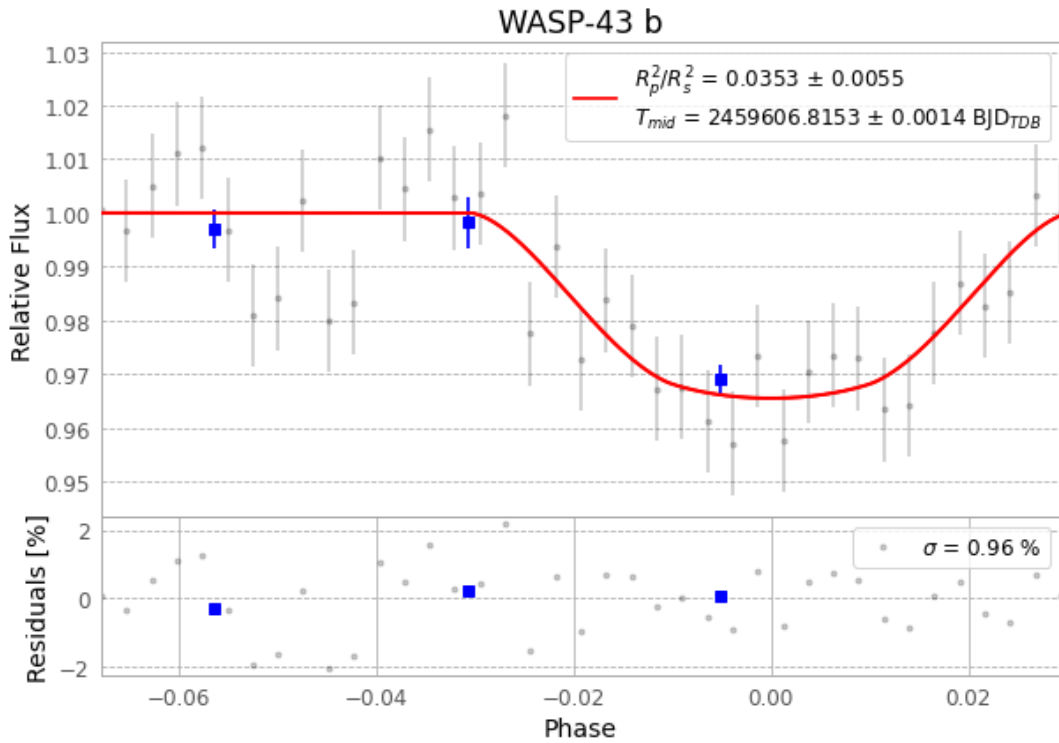


Figure 1 — FinalLightCurve_WASP-43 b_2022-01-26.png (SHA-256 f06d6f3c6c2c2872...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7027c71528e6bc57...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

39 FITS frames (26 MB), WASP-43220127060612.FITS → WASP-43220127080014.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220127.log (e06607d6a68c...), FinalLightCurve_WASP-43 b_2022-01-26.png (f06d6f3c6c2c...), FinalLightCurve_WASP-43 b_2022-01-26.csv (fdb4674b1a1e...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 08:35Z.